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Leveraging commitment- and collaboration-oriented human resource management for business model innovation in small-and-medium-sized enterprises: A dual-path perspective of knowledge acquisition and application

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Abstract

This study investigates the impact of commitment- and collaboration-oriented human resource management (HRM) systems on business model innovation (BMI) in small and medium-sized enterprises (SMEs), drawing on the knowledge-based view (KBV) as well as HRM and BMI literature. The findings reveal that commitment-oriented HRM systems, which focus on fostering internal trust and collaboration, enhance knowledge acquisition efficiency, thereby positively influencing BMI. Conversely, collaboration-oriented HRM systems, designed to facilitate external partnerships and boundary-spanning activities, improve knowledge application, further driving BMI outcomes. Using survey data from 192 SMEs in China, the study provides empirical validation of these relationships. Furthermore, the results show that BMI significantly enhances SMEs' innovation performance, particularly in new product development. The research also identifies the moderating role of flow-facilitating HRM practices, which strengthen the relationship between BMI and innovation performance by promoting effective knowledge flows across the organization. These findings offer both theoretical and practical implications, demonstrating how SMEs can strategically employ HRM systems to optimize knowledge management processes and achieve sustained innovation success.

Keywords Business model innovation · HRM systems · Knowledge management · Innovation performance · SMEs

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A substantial body of research has examined the effects of human resource management (HRM) systems on firms' financial and innovation performance (e.g., Ho & Kuvaas, 2020; Lee et al., 2019; Schmidt et al., 2018; Zhou et al., 2013). These studies rest on several key assumptions. First, HRM systems are believed to foster innovation by enhancing employee ownership and engagement (Azevedo et al., 2021; Alfes et al., 2013; Degbey et al., 2021). Second, they enhance employees' creativity and innovation capabilities (Jiang, Lepak et al., 2012; Jiménez-Jiménez & Sanz-Valle, 2008; Renkema et al., 2022). Third, HRM systems help shape a supportive organizational culture and climate (Datta et al., 2023; Lau & Ngo, 2004; Heffernan et al., 2009). While these assumptions have been well articulated and empirically validated, persistent questions remain regarding the mechanisms through which HRM systems influence innovation (Seeck & Diehl, 2017; Vrontis et al., 2022).

Recent research suggests that knowledge management may serve as a critical mediating mechanism in the relationship between HRM systems and innovation outcomes (Seeck & Diehl, 2017). For example, Chen and Huang (2009) found that knowledge management processes mediate the influence of HRM systems on innovation performance. Supporting this view, studies highlight the importance of enabling firms to create, disseminate, and integrate knowledge effectively to achieve innovation (Bhatti et al., 2021; Jiménez-Jiménez & Sanz-Valle, 2012; Kraus et al., 2023). The knowledge-based view (KBV) of the firm provides a compelling theoretical lens for understanding this relationship. KBV posits that knowledge is a firm's most critical strategic resource, serving as the foundation for innovation and long-term competitive advantage (Grant, 1996). From this perspective, HRM systems play a vital role in fostering the acquisition, dissemination, and application of knowledge, enabling firms to innovate successfully in dynamic environments (Durst et al., 2023).

Although the relationship between HRM systems and knowledge management has been widely acknowledged, limited research has explored the specific knowledge management processes involved, particularly in the context of business model innovation (BMI) (Seeck & Diehl, 2017; Spanellis et al., 2021; Papa et al., 2020). BMI is a distinct form of organizational innovation that focuses on rethinking and redesigning how a firm creates, delivers, and captures value (Codini et al., 2023). Unlike product or process innovations, BMI fundamentally transforms a firm's operational framework, requiring firms to effectively manage and leverage knowledge resources to remain competitive (Clauss, 2017).

The knowledge management literature emphasizes that knowledge acquisition efficiency and knowledge application are essential for a firm's ability to innovate through BMI (Adomako, 2024; Bashir & Farooq, 2019; Malhotra, 2000). Knowledge acquisition efficiency refers to the firm's capacity to systematically gather, process, and internalize knowledge from both internal and external sources, ensuring the efficient generation of actionable insights with minimal resources (Shin & Pérez-Nordtvedt, 2020). In contrast, knowledge application involves effectively utilizing acquired knowledge by adapting and integrating it into organizational processes, practices, and decision-making to create value and drive innovation (Ode & Ayavoo, 2020). By enabling firms to gather, integrate, and apply knowledge, these two processes collectively ensure that knowledge is not only acquired but also effectively applied to achieve innovation outcomes and support business model redesign (Chen et al., 2022; Ji et al., 2021; Turulja & Bajgorić, 2020).

Commitment-oriented HRM systems enhance knowledge acquisition efficiency by fostering trust, collaboration, and alignment within the organization, creating an environment where employees are motivated to acquire knowledge from internal teams and external stakeholders such as customers and experts (Hauff et al., 2014; Zhou et al., 2013). These systems cultivate a culture of collaboration and knowledge sharing, enabling firms to identify, process, and integrate diverse knowledge inputs into their organizational processes, thereby building a strong foundation for innovation and business model redesign (Chiang et al., 2011; Cera et al., 2024). Rather than focusing on how knowledge is acquired, collaboration-oriented HRM systems emphasize how external knowledge is adapted and applied to drive innovation outcomes in dynamic environments. These systems prioritize trust-based external partnerships and cross-boundary collaboration, ensuring that external knowledge aligns with organizational goals and is effectively integrated into innovation processes (Hong et al., 2019; Zhou et al., 2013). Together, these HRM systems complement each other: while commitment-oriented systems focus on acquiring and integrating knowledge, collaboration-oriented systems ensure its effective application, providing firms with a holistic approach to managing knowledge resources and supporting BMI.

Building on the KBV and knowledge management perspectives, this study seeks to illuminate the role of HRM systems in enabling knowledge management processes that support BMI. Specifically, we focus on two types of HRM systems, namely commitment-oriented HRM systems and collaboration-oriented HRM systems, which have been shown to influence organizational innovation outcomes in distinct ways (Zhou et al., 2013). Commitment-oriented HRM systems emphasize fostering internal cohesion and employee alignment, facilitating the efficient acquisition of knowledge within the organization (Collins & Smith, 2006; Methot et al., 2018; Salas-Vallina et al., 2021). In contrast, collaboration-oriented HRM systems focus on external engagement and adaptability, enabling firms to apply external knowledge efficiently to drive innovation (Gerhart & Feng, 2021; Powell et al., 1996). While previous studies often treat the impact of HRM systems on organizational knowledge management as uniform, the potentially distinct contributions of these systems to BMI remain poorly understood.

This study makes three key contributions to the literature by linking knowledge management concepts with HRM practices in the context of BMI. First, it advances the theoretical understanding of BMI by showing how HRM systems enable the knowledge flows required for business model transformation. Commitment-oriented HRM systems enhance knowledge acquisition by fostering trust and collaboration, while collaboration-oriented HRM systems enable knowledge application by facilitating external engagement and adaptation of external knowledge. Second, it provides empirical evidence of the differentiated and complementary roles of these HRM systems in addressing BMI's dual demands. Commitment-oriented systems strengthen internal capabilities for knowledge acquisition, while collaboration-oriented systems ensure external adaptability for knowledge application. Third, it introduces flow-facilitating HRM practices as key enablers of BMI implementation. These practices, such as communication and cross-functional coordination, ensure seamless knowledge flows and help translate BMI strategies into tangible innovation outcomes. Together, these contributions deepen our understanding of how HRM systems and practices support both the design and implementation of BMI.

Figure 1 summarizes the conceptual model of this study, illustrating the theoretical pathways and relationships explored in our analysis.

Theoretical background and hypotheses development

Knowledge-based view

The KBV of the firm posits that knowledge is the most critical strategic resource for achieving and sustaining competitive advantage (Grant, 1996). Extending the RBV (Wernerfelt, 1984; Barney, 1991), KBV emphasizes that firms exist to create, transfer, and apply knowledge effectively, as knowledge underpins innovation and performance. Unlike tangible resources, knowledge is dynamic, appreciating through use and integration, making it especially pivotal for firms in competitive and rapidly changing environments (Kogut & Zander, 1992). For small and medium-sized enterprises (SMEs), which often operate with resource constraints, managing and leveraging knowledge effectively is essential for survival and differentiation in the marketplace (Durst et al., 2023).

KBV is closely intertwined with knowledge management, which encompasses the processes of acquiring, disseminating, and applying knowledge to enhance organizational performance and innovation (Pereira & Bamel, 2021; Grant, 1996; Wiig, 1997). In this study, commitment-oriented HRM systems are proposed to enhance

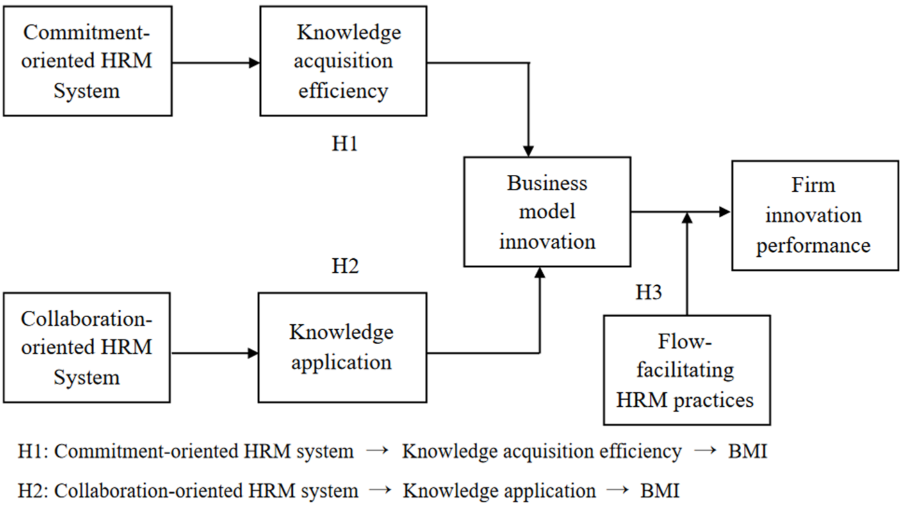


Fig. 1 Conceptual framework for the study

knowledge acquisition efficiency, facilitating the effective gathering and internalization of knowledge (Yu et al., 2022; Zhou et al., 2013), while collaboration-oriented HRM systems support the application of external knowledge to drive innovation. These processes are central to enabling BMI, a knowledge-intensive activity where firms redesign how they create, deliver, and capture value (Zhou et al., 2013). Furthermore, flow-facilitating HRM practices ensure smooth knowledge flow across the organization, maximizing the impact of BMI on innovation performance. Thus, KBV provides a robust theoretical framework for understanding the interplay between HRM systems, knowledge management, BMI, and innovation performance, particularly in the context of SMEs.

Knowledge management through HRM systems

The relationship between HRM systems and knowledge management lies in the ability of HRM systems to enhance a firm's capacity to generate, manage, and disseminate knowledge by fostering employees' learning capabilities and collaboration (Gope et al. 2018; Lei et al. 2021a, b; Malik et al. 2020; Yahya and Goh 2002). Research consistently underscores the critical role of HRM systems in shaping employees' skills, attitudes, and behaviors, which are essential for driving knowledge creation, sharing, and application within firms (Pham et al., 2023; Podolsky & Hackett, 2023; Saks, 2022). These systems go beyond routine operational tasks, acting as strategic mechanisms that cultivate a learning-oriented environment and encourage continuous knowledge sharing. By aligning human capital with organizational goals, HRM systems create fertile conditions for innovation and secure competitive advantage through effective knowledge practices (Lei et al. 2021a, b; Malik et al. 2020). This alignment ensures that employees are not only well-prepared but also actively engaged in the collective mission of knowledge cultivation, which is critical for long-term organizational success.

Knowledge management is a dynamic and multifaceted process that enhances organizational capabilities and performance, especially in the context of innovation (Pellegrini et al., 2020; Shahzad et al., 2020). It encompasses several interconnected phases: knowledge acquisition, which involves identifying, gathering, and organizing valuable insights and resources (Obeidat et al., 2016); knowledge dissemination, which involves sharing and distributing knowledge across the organization; and knowledge application, which refers to the utilization of acquired knowledge to strengthen capabilities and gain competitive advantages (Ode & Ayavoo, 2020; Wu & Chen, 2014). This study focuses on knowledge acquisition efficiency, which refers to acquiring relevant knowledge quickly and with minimal resource use (e.g., time, cost, or effort), and knowledge application, which emphasizes the efficient deployment of knowledge to support BMI.

The emphasis on efficiency is particularly relevant to SMEs, given their limited resources and the need to respond rapidly to changing market conditions. Efficient knowledge acquisition enables SMEs to gather actionable insights without overextending their limited time, financial resources, or personnel. For SMEs, delays or inefficiencies in acquiring knowledge can result in missed opportunities or reduced competitiveness. By acquiring knowledge efficiently, SMEs can act promptly and make informed decisions, which is critical for sustaining agility and competitiveness in dynamic environments (Lei et al. 2021a, b; Pellegrini et al. 2020).

Similarly, knowledge application in this study focuses on the efficient integration and use of knowledge to drive innovation outcomes. Efficient knowledge application enables SMEs to quickly adapt and apply knowledge to enhance business processes, products, or services while optimizing resource use. This focus on efficiency enables SMEs to maintain agility and flexibility, which are essential for seizing opportunities and implementing BMI in fast-changing markets. Efficient application enables SMEs to maximize the value of acquired knowledge while avoiding the costs and time delays associated with resource-intensive implementation processes.

Within this framework, effective HRM systems can act as enablers of knowledge management by providing structure, clarity, and consistency in the processes of acquiring and applying knowledge. Effective HRM systems serve as communication networks that convey strategic objectives and behavioral expectations to employees, fostering a shared understanding of organizational goals. This alignment reduces inefficiencies caused by miscommunication and helps shape an HRM climate conducive to learning and knowledge sharing (Hauff et al., 2017). Conversely, weak HRM systems can lead to misalignments and inefficiencies, undermining the success of knowledge management initiatives.

Commitment-and collaboration-oriented HRM systems

HRM systems play a vital role in shaping organizational capabilities and driving innovation (Park et al., 2019; Sardi et al., 2021). The literature identifies various HRM systems, including control-oriented, commitment-oriented, collaboration-oriented, and relational HRM systems (Hauff et al., 2014; Zhou et al., 2013; Bannya et al., 2023). Among these, commitment-oriented and collaboration-oriented HRM systems are particularly important for enhancing knowledge management processes and fostering BMI (Chin et al., 2025; Loon et al., 2020; Park et al., 2019; Zhou et al., 2013). Commitment-oriented HRM systems focus on developing internal capabilities and creating a collaborative work environment, whereas collaboration-oriented HRM systems emphasize building external partnerships and utilizing external knowledge sources (Hauff et al., 2014; Zhou et al., 2013). These systems complement each other by addressing different aspects of knowledge management, enabling organizations to effectively acquire and apply knowledge to sustain innovation.

Commitment-oriented HRM systems are designed to enhance employee development, long-term engagement, and trust (Zhou et al., 2013). These systems adopt a high-investment approach, prioritizing practices that build employee skills, autonomy, and loyalty (Lawler, 1992; Walton, 1985). By fostering collaboration, shared goals, and psychological safety, commitment-oriented HRM systems create an environment in which knowledge can be efficiently acquired, shared, and integrated (Park et al., 2019; Roche, 1999). Practices such as employment security and open communication encourage employees to freely exchange ideas and expertise across teams, while initiatives such as job rotation and cross-departmental training broaden employees' exposure to diverse knowledge inputs and organizational processes.

Commitment-oriented HRM systems primarily enhance knowledge acquisition efficiency, defined as the organization's ability to acquire and process knowledge from internal and external sources, including employees, customers, suppliers, and other relevant stakeholders (Bloodgood, 2019; Grimpe & Kaiser, 2010). These systems cultivate a culture of trust, collaboration, and shared purpose, enabling employees to effectively absorb and integrate knowledge from various sources into organizational processes. By equipping employees with the skills and capabilities needed to identify and capture actionable insights, commitment-oriented HRM systems strengthen the firm's ability to leverage knowledge for innovation and growth. While commitment-oriented HRM systems excel at fostering knowledge acquisition, they are less focused on external engagement and the broader application of knowledge in dynamic, networked environments (Ishak et al., 2023).

Collaboration-oriented HRM systems, on the other hand, focus on external engagement and the development of strategic relationships with external stakeholders, such as customers, suppliers, academic institutions, and industry partners (Zhou et al., 2013; Hong et al., 2019). These systems prioritize practices that facilitate the acquisition and application of knowledge from external sources, enabling organizations to benefit from diverse perspectives and expertise. Examples of such practices include cross-organizational collaborations, external training programs, and co-development initiatives that enable firms to access and integrate external knowledge into their innovation processes (Collins & Clark, 2003; Powell et al., 1996).

The primary contribution of collaboration-oriented HRM systems lies in enhancing knowledge application, which refers to the firm's capacity to effectively integrate and utilize knowledge from external sources to address opportunities, solve challenges, and drive innovation (Ode & Ayavoo, 2020; Song et al., 2005). These systems help firms overcome barriers such as cultural mismatches and coordination inefficiencies, enabling seamless knowledge transfer and collaboration with external stakeholders. For instance, partnerships with academic institutions can provide access to cutting-edge research, while collaborations with suppliers and customers may reveal valuable insights into emerging market trends and technologies (Zhou et al., 2013; Hong et al., 2019). By fostering trust-based relationships, collaboration-oriented HRM systems ensure that external knowledge is effectively adapted to the organization's strategic goals and innovation processes, enabling firms to remain competitive in dynamic environments.

Although both HRM systems may influence multiple aspects of knowledge management, this study focuses on the dominant pathways through which they exert their primary effects: commitment-oriented HRM systems enhancing knowledge acquisition and collaboration-oriented HRM systems improving knowledge application. This theoretical framing is grounded in the distinct focuses and design of these systems. Commitment-oriented HRM systems are particularly effective at creating a cohesive, collaborative internal environment, which is essential for acquiring and integrating knowledge from employees, customers, suppliers, and other close stakeholders (Hauff et al., 2014; Park et al., 2019; Zhou et al., 2013). These systems emphasize practices such as open communication, internal training, and job rotation, which collectively foster an environment of trust and collaboration that builds the organization's capacity to acquire knowledge efficiently (Nieves & Osorio, 2017).

While commitment-oriented HRM systems may indirectly contribute to knowledge application, their primary design focuses on strengthening organizational learning and knowledge flows rather than on the external engagement required to apply diverse external knowledge.

Similarly, collaboration-oriented HRM systems are outward-focused and designed to enhance the firm's ability to integrate and apply external knowledge. These systems emphasize boundary-spanning practices and relational trust, enabling firms to integrate external knowledge into their innovation processes (Hong et al., 2019; Zhou et al., 2013). For example, practices such as co-development with external partners and participation in external training programs provide direct channels for identifying, integrating, and applying external knowledge. While collaboration-oriented HRM systems may indirectly influence knowledge acquisition by exposing firms to new ideas and information, their primary contribution lies in enabling the effective use of external knowledge to address organizational challenges and drive innovation.

The focus on these dominant pathways is consistent with prior research, which highlights the unique strengths of commitment- and collaboration-oriented HRM systems in supporting specific knowledge management outcomes (Zhou et al., 2013; Seeck & Diehl, 2017). Including additional cross-path relationships, such as the potential influence of commitment-oriented HRM systems on knowledge application or collaboration-oriented HRM systems on knowledge acquisition, would add unnecessary complexity to the theoretical model without significantly enhancing its explanatory power. By focusing on the most theoretically grounded relationships, this study provides a clear and parsimonious framework for understanding the role of HRM systems in fostering BMI. Together, these systems complement one another, addressing distinct aspects of knowledge management while supporting the KBV, which highlights the importance of both internal knowledge integration and external knowledge adaptation for achieving competitive advantage (Grant, 1996). By fostering both internal and external knowledge flows, these HRM systems enable organizations to better leverage their knowledge resources, driving innovation and achieving sustainable competitive advantage (Pellegrini et al., 2020). While commitment-oriented systems ensure a strong foundation for organizational learning, collaboration-oriented systems enable firms to adapt to market demands and technological advancements, fostering agility and sustained innovation performance (Cheng & Shiu, 2021; Frynas et al., 2018).

Internal commitment-oriented HRM systems and BMI

Commitment-oriented HRM systems enable SMEs to achieve high levels of knowledge acquisition efficiency, which is critical for fostering BMI. Knowledge acquisition efficiency refers to the organization's ability to effectively acquire, process, and integrate diverse knowledge inputs to support innovation and value creation (Bloodgood, 2019; Shin & Pérez-Nordtvedt, 2020; Yli-Renko et al., 2001). This construct encompasses the firm's capacity to leverage knowledge from various sources, whether through internal collaboration or engagement with external stakeholders, to ensure continuous learning and adaptability (Grimpe & Kaiser, 2010; López-Sáez et al., 2010). Commitment-oriented HRM systems play a central role in enhancing this efficiency by fostering a workforce that is highly skilled, motivated, and capable of engaging in collaborative knowledge-driven activities (Yu et al., 2022). These systems are designed to develop employees' potential, build trust, and create a culture that facilitates the seamless flow of knowledge across organizational boundaries (Kim & Shin, 2019). Commitment-oriented HRM systems enhance knowledge acquisition efficiency by strengthening organizational processes that support knowledge sharing and integration. They enable firms to establish multiple mechanisms for acquiring and utilizing knowledge that foster innovation.

First, commitment-oriented HRM systems promote knowledge sharing and integration within the organization by implementing HRM practices that encourage employees to exchange information and collaborate across departments. These systems establish efficient communication platforms that facilitate knowledge sharing and collaboration among employees, thereby supporting organizational learning and fostering an environment for collective knowledge creation (Wright & Essman, 2021). Employees are encouraged to actively participate in identifying and integrating diverse knowledge sources (Lei et al. 2021a, b). Furthermore, these HRM systems create structured processes that ensure the right knowledge is delivered to the right people at the right time, forming a robust knowledge-processing system that transforms tacit knowledge into explicit knowledge and vice versa (Katou et al., 2021). The result is a dynamic knowledge-sharing culture that drives innovation and value creation (Castaneda & Cuellar, 2020).

Second, commitment-oriented HRM systems foster organizational citizenship behavior, which enhances structural connectedness and collaboration among employees and departments. By cultivating a work environment that values teamwork, community building, and interconnectedness, these systems create a climate of mutual respect and trust, enabling employees to establish broader interpersonal connections (Remneland et al., 2022; Cooper et al., 2019). Additionally, commitment-oriented HRM systems develop and sustain an organizational knowledge language, making knowledge more accessible, transferable, and actionable across the firm (Hock-Doepgen et al., 2021). This continuous development of the knowledge base ensures that employees can refresh and adapt their knowledge to meet emerging challenges and opportunities (Valentim et al., 2016).

The importance of knowledge acquisition efficiency lies in its ability to generate multiple BMI possibilities. According to the literature, BMI requires firms to internalize knowledge configurations and leverage knowledge effectively during value-creation processes (Amit & Zott, 2001). High knowledge acquisition efficiency enables firms to identify complementary knowledge and resources within and across organizational boundaries, thereby unlocking synergies that drive innovation (Joseph & Wilson, 2018). For example, interactions among divisional managers or cross-functional teams enhance organizational coupling, enabling firms to access complementary knowledge streams and resources for innovative activities (Ocasio et al., 2018). Research on knowledge complementarity highlights that both tacit and explicit knowledge are essential for BMI (e.g., Gamble, 2020; Magnier-Watanabe & Benton, 2017). For instance, marketing departments may provide insights into customer behavior, while operational teams may share knowledge about internal inefficiencies, enabling the design of more effective business models (Visnjic et al., 2016).

Commitment-oriented HRM systems also enhance knowledge acquisition efficiency by enabling the firm to engage effectively with external stakeholders (Cera et al., 2024; Kim & Shin, 2019), such as key customers and product development experts. These relationships provide access to valuable knowledge that can refine and enhance business models (Yi et al., 2022). For example, insights from customers help improve customer satisfaction by tailoring the business model to meet specific needs, while interactions with industry experts provide foresight into technological and industry trends (Ocasio et al., 2018). Commitment-oriented HRM systems equip employees with the skills and capabilities to integrate these diverse knowledge sources, ensuring that firms can continuously refresh their knowledge base and adapt to changing environments (Hauff et al., 2014; Roche, 1999; Zhou et al., 2013). This cyclical process mitigates the risks of knowledge obsolescence while driving long-term BMI success.

Taken together, we propose the following hypothesis:

Hypothesis 1 Knowledge acquisition efficiency mediates the positive relationship between commitment-oriented HRM systems and BMI in SMEs.

External collaboration-oriented HRM systems and BMI

External collaboration-oriented HRM systems differ from internal commitment-oriented HRM systems in their emphasis on fostering strategic relationships with external stakeholders, such as customers, suppliers, academic institutions, and industry partners (Zhou et al., 2013). While commitment-oriented systems are essential for creating a strong internal foundation for knowledge acquisition and integration, collaboration-oriented systems are uniquely designed to facilitate the application of external knowledge, enabling organizations to leverage insights and expertise from their external environment to BMI.

These HRM systems enable firms to establish, nurture, and maintain high-quality, trust-based relationships with a wide range of external stakeholders (Hong et al., 2019; Zhou et al., 2013). Through structured mechanisms, collaboration-oriented HRM systems ensure that valuable external knowledge is identified, captured, and integrated into organizational processes. For instance, customer partnerships provide insights into shifting market demands, while collaborations with suppliers or industry experts introduce novel technologies and operational efficiencies. Such external engagement mechanisms ensure that organizations remain agile and responsive to changing environments, laying the foundation for sustained innovation (Borisov & Vinogradov, 2019). While internal HRM systems may indirectly support external engagement, collaboration-oriented HRM systems excel in promoting reciprocal communication and long-term partnerships that foster the exchange and application of external knowledge. By building relational trust and aligning with external stakeholders, these systems facilitate critical knowledge flows for collaborative decision-making and the seamless integration of external insights into innovation activities. This makes collaboration-oriented HRM systems especially valuable for SMEs, which often rely on external networks to compensate for resource limitations.

Knowledge application, a vital process in knowledge management, involves integrating and utilizing external knowledge to address challenges, seize opportunities, and foster innovation (Ode & Ayavoo, 2020; Song et al., 2005). Unlike knowledge acquisition, which focuses on gathering and organizing information, knowledge application emphasizes transforming external knowledge into actionable insights that create value and sustain competitive advantage (Choi et al., 2010; Jarrahi et al., 2023). Collaboration-oriented HRM systems facilitate this process by providing employees with the tools, practices, and opportunities to apply external knowledge effectively. For example, collaboration-oriented HRM systems often promote alliances, partnerships, and co-development initiatives that allow firms to capitalize on complementary expertise and resources (Remneland et al., 2023; Zhou et al., 2013). Collaborations with academic institutions may provide access to cutting-edge research, while partnerships with industry professionals can reveal emerging market trends or technical advancements. These systems create an environment of mutual exchange, enabling firms to apply external insights directly to BMI processes, such as designing new customer offerings or optimizing value chains (Seeck & Diehl, 2017).

Collaboration-oriented HRM systems also play a pivotal role in overcoming barriers to the application of external knowledge, such as communication mismatches, cultural differences, and coordination inefficiencies. By fostering trust, relational capital, and effective coordination mechanisms, these HRM systems ensure that external knowledge flows seamlessly into the organization and is applied to innovation initiatives (Chesbrough, 2010). This ability to access and apply external expertise is especially critical for SMEs, which often lack the internal resources to achieve innovation goals independently. Moreover, collaboration-oriented HRM systems enhance employees' motivation, abilities, and opportunities (AMO framework) to participate in external collaborations and apply external knowledge effectively (Greer & Stevens, 2015; Hong et al., 2019). Practices such as external training programs, knowledge-sharing forums, and stakeholder engagement initiatives empower employees to integrate external insights into BMI processes. These systems actively encourage employees to work with external partners, identify actionable insights, and turn them into innovative solutions.

By fostering the application of external knowledge, collaboration-oriented HRM systems enable SMEs to continuously adapt to dynamic market conditions, identify emerging opportunities, and address external challenges. Leveraging external knowledge supports both exploration (discovering new opportunities) and exploitation (optimizing existing capabilities), resulting in both breakthrough and incremental innovations that strengthen BMI (Du, 2021; Inigo et al., 2017). This adaptability is essential for SMEs seeking to maintain their competitive edge in today's fast-paced, interconnected business environment (Yi et al., 2022).

Taken together, collaboration-oriented HRM systems serve as a vital mechanism for integrating and applying external knowledge, enabling SMEs to overcome resource constraints and innovate their business models. Therefore, we propose the following hypothesis:

Hypothesis 2 Knowledge application mediates the positive relationship between collaboration-oriented HRM systems and BMI in SMEs.

The moderating role of flow-facilitating HRM practices

BMI are pivotal for firms' innovation performance, driving continuous innovation in their business strategies (Latifi et al., 2021; Vrontis et al., 2022). From the KBV, BMI can be seen as a dynamic reconfiguration of a firm's knowledge resources to support new approaches to value creation, delivery, and capture. BMI enhances innovation performance by enabling firms to integrate and utilize knowledge in novel ways, such as through reconfigurations of activities, processes, and technologies (Andreini et al., 2022; Spieth et al., 2014). However, while BMI provides the structural foundation for innovation, its successful implementation relies heavily on the firm's ability to manage and apply knowledge effectively. Flow-facilitating HRM practices, rooted in knowledge management principles, are essential for this implementation as they enable the seamless flow, exchange, and application of knowledge within the firm. Thus, we propose that flow-facilitating HRM practices moderate the relationship between BMI and innovation performance by ensuring that the knowledge resources embedded in BMI are effectively mobilized and applied throughout the organization. Specifically, effective BMI implementation requires practices that establish new communication channels, implement innovative reward systems, and expand decision-making delegation, all of which support knowledge flows and enhance organizational agility.

There are two primary reasons supporting the interactive effect of BMI and flow-facilitating HRM practices on innovation performance from the perspectives of the KBV and knowledge management. First, flow-facilitating HRM practices create a robust knowledge-sharing framework that is critical for implementing BMI. From the KBV perspective, knowledge is the most strategic resource of the firm, and its effective mobilization is essential for innovation. Flow-facilitating HRM practices help firms establish communication systems that ensure the integration and dissemination of both explicit and tacit knowledge generated during BMI processes (Sung & Choi, 2018). By fostering open communication and reducing knowledge silos, these practices allow employees to actively engage in BMI initiatives, leveraging their expertise to refine and implement new business models. From a knowledge management perspective, these practices enhance the efficiency of knowledge flows by empowering employees to acquire, share, and apply BMI-related knowledge. This empowerment leads to greater implementation efficiency, reducing production costs and optimizing resource utilization (Alfes et al., 2013). Moreover, enabling employees to share and refine knowledge supports the iterative improvement of products and services, allowing firms to deliver enhanced value to external stakeholders, such as customers, suppliers, and collaborators (Shipton et al., 2017). By aligning HRM practices with BMI objectives, firms ensure that knowledge resources are fully utilized to achieve innovation goals.

Second, the synergy between BMI and flow-facilitating HRM practices fosters an innovative and opportunity-seeking organizational culture, which is essential for sustained innovation. From the KBV, this synergy allows firms to maximize the potential of their knowledge resources by creating an environment where employees can explore, experiment, and implement new ideas. Flow-facilitating HRM practices provide employees with the tools, autonomy, and incentives needed to actively contribute to BMI (Jiang et al., 2012; Lin & Sanders, 2017). These practices also ensure that the firm's business logic for value creation, delivery, and capture is communicated effectively across all levels of the organization, aligning employees with BMI's strategic vision. From a knowledge management perspective, flow-facilitating HRM practices enhance the firm's ability to integrate diverse knowledge inputs and facilitate collaboration across organizational boundaries. This enables the firm to uncover new product and service innovation opportunities while aligning them with its broader innovation strategy. By fostering an environment where knowledge is continuously shared and applied, flow-facilitating HRM practices enhance the firm's capacity to implement BMI effectively, thereby improving innovation performance.

Based on these insights, we propose the following hypothesis:

Hypothesis 3 Flow-facilitating HRM practices moderate the relationship between BMI and innovation performance in SMEs.

Research design

Sample and data collection

We focused our study on the high-tech manufacturing industry in China due to the government's strategic emphasis on innovation as a crucial lever for rapid economic development. Encouragements and incentives are provided for firms to foster innovation and maintain competitiveness (Pan et al., 2022; Zhang et al., 2023). In 2021, we conducted a survey of SMEs in the Zhongguancun National Innovation Demonstration Zone in Beijing to test our proposed hypotheses. Known as China's most innovative science park, Zhongguancun, along with Beijing's ranking as one of the top four global startup ecosystems in 2021, epitomizes the dynamic environment ideal for our study. The Chinese Government's "Made in China 2025" initiative plays a significant role in this context by stimulating SMEs to refine their business models and adopt emerging technologies. This initiative is geared toward transforming SMEs not only to bolster China's status as a manufacturing giant but also to elevate it to a world leader in sustained manufacturing innovation.

We collected data from SMEs across both high-technology and non-high-technology sectors. To enhance response rates and ensure the relevance of our survey content, we initially conducted a pretest with senior executives from 15 selected SMEs. This preliminary test aimed to confirm the realism, clarity, and appropriateness of the questions for the Chinese SME environment. Based on the feedback and suggestions from these executives, we refined the survey instruments to ensure they were understandable and relevant to our target audience. From July to August 2021, our research team and field assistants distributed the revised survey questionnaires to potential SMEs situated in various innovation and industrial parks within the expansive Zhongguancun National Innovation Demonstration Zone.

To determine our sample, we obtained a comprehensive list of 8,124 SMEs from the Ministry of Science and Technology, focusing specifically on those located within the expansive Zhongguancun National Innovation Demonstration Zone. Of these, 4,346 SMEs were identified as potential participants based on their geographical location. We further refined our selection by confirming that 3,672 of these enterprises had been operational for five years or more, a criterion that ensured they had sufficient experience and stability for our study. We then contacted these established firms to gauge their willingness to participate in our survey.

During this process, we also verified whether they met several critical selection criteria: the implementation of commitment- and collaboration-based HRM systems for at least the past three years, active engagement in innovative activities during this period, and the availability and willingness of their senior executives (owners, CEOs, or COOs, and HR Managing Directors) to participate in the survey. A total of 593 SMEs confirmed their intent to participate, signaling robust interest in our research objectives.

To improve response rates and ensure thorough participation, we adopted a multi-step communication strategy. Initially, we sent out an invitation letter that included not only the full questionnaire but also a detailed cover letter explaining the research purpose and highlighting its potential contributions to societal and economic advancements. This initial communication was followed two weeks later by a series of reminder messages, complemented by strategic follow-up telephone calls to those who had not yet responded. By the August 2021 deadline, we had received completed questionnaires from 247 firms. However, our quality control measures required us to exclude 22 firms that had been in operation for less than five years and an additional 13 that had not engaged in any innovation activities. This scrutiny led us to a refined cohort of 212 eligible responses that met all our rigorous selection criteria. Of these, 192 were validated as complete and accurate, forming the robust dataset that we used to test our proposed hypotheses. This methodical approach not only ensured the reliability of our data but also underscored our commitment to drawing meaningful and applicable insights from a well-defined sample of Chinese SMEs.

The sample of this study comprises a diverse array of SMEs. The SMEs range in age from 5 to 26 years, representing both relatively new enterprises and well-established ones. Employee numbers among these SMEs vary from 12 to 243, offering insights into operational and management challenges across different business sizes. Additionally, the founding teams of these SMEs consist of between 1 and 6 members. Industry representation in the sample is strategically divided, with 56.3% of the firms engaged in high-technology sectors, including artificial intelligence, telecommunications, robotics, and both software and hardware development. The remaining 43.7% of firms come from non-high-technology sectors such as the service industry, food production, and agriculture.

Measurement

Dependent variables

We measured BMI using a scale with night items across the three dimensions recommended by Heider et al. (2021): value creation (4 items), value proposition (3 items), and value capture (2 items). All items were measured using a 7-point Likert scale. We measured innovation performance using two alternative ways. First, following the literature, we used the count of new product introductions over the past five years, including new products introduced to the market, the firm, and the world. Second, we used three scale items (Frishammar & Åke Hörte, 2005); for example, we asked the CEOs to indicate which of the following statements best suited their company: 1 = Changes in products/services have been mostly minor; 7 = Changes in products/services have usually been dramatic. Items were measured using a 7-point Likert scale.

Independent variables

Following the literature (e.g., Zhou et al., 2013), we measured commitment-oriented HRM systems using a 15-item bundle that includes overarching goal setting, self-managed teamwork, information sharing, and an employee proposal mechanism. Those items were measured using a 7-point Likert scale. A collaboration-oriented HRM system was measured using a scale based on existing studies, including six items such as a formal external learning program with business partners and building extensive social networks (e.g., Lepak et al., 2003; Zhou et al., 2013), with responses drawn from Directors of Human Resources. Those items were measured using a 7-point Likert scale.

Mediating and moderating variables

Our framework includes two mediating variables: knowledge acquisition efficiency and knowledge application. We measured knowledge acquisition efficiency using eight items across two dimensions: efficiency in knowledge acquisition from main customers and from product development experts (Shin & Pérez-Nordtvedt, 2020). We measured knowledge application by using five items developed by Song et al. (2005). We captured flow-facilitating HRM practices, the moderating variable, using a three-item scale from existing studies (e.g., Sung & Choi, 2018). Those items were measured using a 7-point Likert scale, with responses drawn from Directors of Human Resources.

Control variables

Our survey gathered extensive data on control variables at the team, firm, and market/industry levels to ensure a comprehensive analysis. These variables were selected for their potential to influence the relationships among HRM systems, BMI, and innovation performance. Controlling for these factors allows us to isolate the effects of the independent variables and ensures that our findings are robust and reliable.

For team-level characteristics, data collection included the highest educational degree attained by CEOs and the total number of founding team members. The CEO's education is a proxy for cognitive and decision-making abilities, which are critical for strategic decisions such as BMI. The number of founding team members reflects the diversity of expertise and perspectives, which can impact team coordination and innovation outcomes. Teamwork capabilities were assessed using a 6-item scale by Brinckmann and Hoegl (2011), with items such as "important information and ideas are shared openly among the members of the founding team," rated on a 7-point Likert scale. Teamwork capabilities are crucial for effective internal communication and collaboration, both of which are essential to knowledge sharing and BMI success. Relational capabilities were evaluated through another 6-item scale by Brinckmann and Hoegl (2011), with items such as "a collaborative atmosphere characterizes the interaction between the founding team and the external marketing partners." Relational capabilities facilitate external knowledge acquisition and partnerships, which are critical for BMI in dynamic environments. Marketing capabilities were measured using a 3-item scale from Zhao et al. (2013), with items such as "the founding team had prior customer service and retention experience in similar industries." Marketing capabilities are included because they help firms understand customer needs and market trends, both of which are important for developing innovative business models. Design capabilities were assessed with a 4-item scale from Zhao et al. (2013), with items such as "the founding team had excellent quality control skills." Design capabilities capture the technical expertise of the founding team, directly affecting product and service quality and supporting innovation. Finally, we also gathered data on the number of startup experiences and industry experience of the founding team members. These variables reflect the team's prior exposure to entrepreneurial activities and industry dynamics, which influence their ability to innovate and implement BMI effectively.

For firm-level characteristics, we collected information on the age of the firm, determined by the number of years since establishment, and firm size, indicated by the number of employees. Firm age is a measure of organizational experience and maturity, which can influence BMI through accumulated knowledge and routines. Firm size reflects the availability of resources and capacity for innovation, with larger firms typically having more resources but potentially facing bureaucratic challenges. Innovation capabilities were gauged by the number of innovation patents granted in the past three years, as patents represent a firm's ability to generate and protect new ideas, which directly impact BMI and innovation performance. Firm growth was evaluated using Zheng's (2012) scale, which captures growth rates in sales, employee numbers, and share value, ranging from 1 (0% growth) to 6 (more than 100% growth). Growth is an indicator of a firm's performance and market success, which can influence its ability to invest in and execute BMI. We also categorized SMEs into high-technology or non-high-technology sectors, specifying the industries involved. Firms in high-technology sectors are often more innovation-driven, while those in non-high-technology sectors may face different pressures and opportunities for BMI.

For market or industrial characteristics, market uncertainty was measured using a three-item scale by Wuyts and Geyskens (2005). Market uncertainty reflects the unpredictability of customer preferences and competition, which can strongly influence the need for BMI. Industrial knowledge intensity was assessed using a four-item scale developed by Martin and Salomon (2003) that captures the extent to which industries rely on specialized knowledge for competitive advantage. Firms in knowledge-intensive industries are more likely to engage in BMI to leverage and apply their expertise effectively. This detailed collection of control variables enables us to comprehensively account for a range of factors that could influence the primary relationships studied in our research, thereby enhancing the robustness and reliability of our findings.

Results

Measurement model

Table 1 presents the correlation matrix among the variables. To assess multicollinearity among all regression variables, we computed variance inflation factors (VIFs). The results revealed a mean value of 1.56, with a range from 1.10 to 3.08, which is well below the cut-off threshold (Petter et al., 2007). These values suggest that there is no multicollinearity among the regression variables. Since the regression variables comprise both formative and reflective constructs, it is necessary to assess their reliability and validity. At the construct level, we assessed the values of Cronbach's alpha (CA). The results showed that the CA values for the variables ranged from 0.834 to 0.965 (see Table 1), all of which were above the threshold of 0.70 (Nunnally, 1978). To examine convergent validity, we assessed the scale's average variance extracted (AVE). Our results indicated that all values exceeded the 0.50 threshold (see Table 1), suggesting satisfactory discriminant validity (Fornell & Larcker, 1981). Therefore, these assessments confirmed the validity of all constructs, supporting the suitability of all scales and the regression analysis.

We examined for common method bias using a Harman one-factor test. This approach assumes that if common method bias is present, either a single factor emerges or one general factor accounts for most of the covariance among the measures (40%) in a factor analysis. The results show that common-method concerns are absent, as nine factors are extracted with eigenvalues greater than 1; no factor accounts for most of the variance; and, taken together, all extracted factors explain 78.77% of the variance. The largest factor identified accounts for 31.92% of the total variance, less than 40%.

Table 1 Correlation matrix, convergent and discriminant valididty of reflective constructs

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
1. Innovation																						
2. Business model innovation	0.465*																					
3. Collaboration oriented-HRM	0.112	0.348*																				
4. Commitment oriented-HRM	0.241*	0.386*	0.105																			
5. CEO educational degree	0.184*	0.378*	0.157*	0.183*																		
6. Firm age	0.177*	0.212*	0.022	0.107	-0.007																	
7. Firm size	0.100	0.138	0.118	0.129	0.171*	0.241*																
8. Founding team member	0.057	-0.082	-0.096	-0.081	-0.101	0.007	-0.117															
9. Start-up experience	0.023	0.061	0.029	0.101	-0.201*	0.069	0.060	0.092														
10. Industry experience	0.094	-0.003	-0.028	0.009	-0.350*	0.050	0.076	-0.009	0.533*													
11. Teamwork capability	0.183*	0.459*	0.190*	0.120	0.332*	-0.015	0.100	-0.051	0.085	-0.035												
12. Relational capability	0.140	0.344*	0.157*	0.143*	0.288*	-0.097	0.009	-0.016	0.026	0.059	0.365*											
13. Marketing capability	0.168*	0.272*	0.159*	0.117	0.120	0.005	0.170*	-0.072	0.091	0.274*	0.226*	0.404*										
14. Design capability	0.120	0.341*	0.173*	0.173*	0.209*	-0.005	0.082	-0.057	0.103	0.174*	0.344*	0.459*	0.448*									
15. Innovative capability	0.418*	0.389*	0.123	0.192*	0.132	0.210*	0.155*	-0.067	0.107	0.166*	0.200*	0.164*	0.234*	0.074								
16. Firm growth	0.226*	0.566*	0.367*	0.255*	0.332*	0.161*	0.195*	-0.169*	0.122	0.048	0.275*	0.315*	0.340*	0.343*	0.311*							
17. Knowledge acquisition efficiency	0.224*	0.353*	0.144*	0.274*	0.249*	0.056	0.040	-0.103	0.059	0.002	0.215*	0.166*	0.141	0.175*	0.082	0.336*						
18. Knowledge application	0.331*	0.398*	0.256*	0.301*	0.224*	0.239*	0.217*	-0.050	0.039	0.050	0.218*	0.263*	0.219*	0.135	0.362*	0.285*	0.253*					
19. Flow-facilitating HRM	0.480*	0.570*	0.071	0.166*	0.148*	0.203*	0.089	0.038	0.021	0.133	0.215*	0.145*	0.139	0.169*	0.438*	0.306*	0.099	0.135				
20. Market uncertainty	-0.053	-0.149*	-0.079	-0.141	-0.201*	0.016	-0.066	0.061	-0.012	0.107	-0.054	0.000	-0.078	-0.123	-0.041	-0.077	-0.050	-0.017	-0.019			
21. Industrial knowledge intensity	0.103	0.275	0.130	0.176*	0.188*	0.028	0.065	-0.105	0.035	0.070	0.190*	0.367*	0.268*	0.342*	0.068	0.307*	0.233*	0.225*	0.242*	0.000		
22. High-technology industry	0.095	-0.139	-0.157*	-0.052	-0.029	-0.064	0.045	0.022	-0.154*	0.045	-0.078	0.002	-0.021	-0.052	0.092	-0.060	-0.020	0.025	0.063	0.035	0.060	
Mean	3.145	4.153	3.942	4.209	3.333	8.635	4.324	3.104	0.516	7.714	4.498	4.178	3.728	4.263	3.385	1.870	3.758	4.041	3.601	4.031	4.329	0.563
Standard deviation	3.435	1.126	0.987	1.005	1.065	3.518	0.747	1.286	1.002	7.242	1.163	1.137	1.400	1.186	3.784	0.804	0.874	1.091	1.080	1.171	1.118	0.497
AVE	n/a	0.761	0.684	0.675	n/a	n/a	n/a	n/a	n/a	n/a	0.834	0.778	0.894	0.847	n/a	n/a	0.694	0.785	0.807	0.720	0.792	n/a
Cronbach's Alpha	n/a	0.961	0.908	0.965	n/a	n/a	n/a	n/a	n/a	n/a	0.960	0.943	0.960	0.940	n/a	n/a	0.938	0.932	0.880	0.834	0.912	n/a

*p>0.05, *AVE* average variance extracted

Baseline tests

Since data were collected from both founder-CEOs (individual, team, and firm levels) and human resource managers, we conducted a hierarchical linear modeling analysis to test the effects of commitment- and collaboration-oriented HRM systems on the BMI of SMEs. We also conducted an OLS analysis, which indicated consistency in the results. Table 2 shows the empirical results of the baseline tests. Model 1 includes individual-, team-, firm-, industry-, and market-level variables. In Model 2, we added the independent variable, commitment-oriented HRM system, to Model 1. The results indicate that a commitment-oriented HRM system is positively and significantly associated with BMI (b = 0.198, p = 0.000). In Model 3, we added another independent variable, a collaboration-oriented HRM system, to Model 1. The empirical results suggest that a collaboration-oriented HRM system is positively and significantly associated with BMI (b = 0.152, p = 0.009).

Finally, these two independent variables, commitment- and collaboration-oriented HRM systems, were simultaneously added into Model 1. The results also indicate positive and significant effects of commitment-oriented HRM (b = 0.199, p = 0.000) and collaboration-oriented HRM (b = 0.153, p = 0.007) on BMI, confirming the results from Models 2 and 3 separately. Therefore, the results of Models 2–4 in Table 2 support the following baselines: (a) Commitment-oriented HRM systems have a positive impact on the BMI of SMEs, and (b) Collaboration-oriented HRM systems have a positive impact on the BMI of SMEs.

Test for mediation

To test the mediation effect of knowledge acquisition efficiency in the relationship between commitment-oriented HRM and BMI, we first take the mediating variable, knowledge acquisition efficiency, as the dependent variable, and then BMI as the dependent variable. Table 3 presents the empirical results. In Models 1 and 2, we tested the effects of a commitment-oriented HRM system on knowledge acquisition efficiency. Model 1 includes the control variables. Model 2 includes both commitment- and collaboration-oriented HRM systems, and the results indicate a positive and significant effect of commitment-oriented (b = 0.163, p = 0.011) on knowledge acquisition efficiency. Thus, we can conclude that a commitment-oriented HRM system has a positive and significant effect on knowledge acquisition efficiency.

In Models 3–6, we took BMI as the independent variable. Model 3 includes all control variables. After that, we added the mediating variable, knowledge acquisition efficiency, as an independent variable in Model 4. The empirical results indicate that knowledge acquisition efficiency has a positive and significant effect on BMI (b = 0.179, p = 0.006). Model 5 includes both the independent variable, the commitment-oriented HRM system, and the mediating variable, knowledge acquisition efficiency. The results indicate that knowledge acquisition efficiency has a positive and significant effect (b = 0.140, p = 0.032) on BMI, as does a commitment-oriented HRM system (b = 0.176, p = 0.002). In Model 6, we also included the independent variable of a collaboration-oriented HRM system. The results are consistent with those of Model 5. In conclusion, the results from Models 1–6 support Hypothesis 1.

Table 2 Results in the effects of commitment- and collaboration HRM on BMI

Variables	Business model innovation			
	Model 1	*Model 2*	Model 3	Model 4
Commitment-oriented HRM		0.198***		0.199***
		0.054		0.053
Collaboration-oriented HRM			0.152**	0.153**
			0.058	0.055
CEO educational degree	0.095	0.082	0.100	0.086
	0.062	0.060	0.061	0.059
Firm age	0.029+	0.026+	0.031+	0.028+
	0.016	0.015	0.016	0.015
Firm size	−0.033	−0.047	−0.045	−0.060
	0.076	0.074	0.075	0.073
Founding team member	−0.016	−0.010	−0.015	−0.008
	0.043	0.041	0.042	0.040
Startup experience	0 0.009	−0.011	0.015	−0.006
	0.066	0.064	0.065	0.063
Industry experience	−0.007	−0.005	−0.006	−0.005
	0.010	0.009	0.009	0.009
Teamwork capability	0.193***	0.199***	0.183**	0.189***
	0.053	0.051	0.052	0.050
Relational capability	0.080	0.075	0.079	0.073
	0 0.059	0.057	0.058	0.056
Marketing capability	0.006	0.010	0.006	0.010
	0.047	0.045	0.046	0.044
Design capability	0.039	0.027	0.036	0.024
	0.056	0.054	0.055	0.053
Innovative capability	0.019	0.013	0.017	0.011
	0.016	0.016	0.016	0.016
Firm growth	0.389***	0.365***	0.329***	0.303***
	0.081	0.079	0.083	0.081
Flow-facilitating HRM	0.392***	0.384***	0.400***	0.392***
	0.058	0.056	0.057	0.055
Market uncertainty	−0.072	−0.056	−0.067	−0.051
	0.047	0.045	0.046	0.045
Industrial knowledge intensity	−0.001	−0.016	−0.004	−0.020
	0.054	0.052	0.053	0.051
High-technology industry	−0.269*	−0.253*	−0.226*	−0.210*
	0.111	0.107	0.110	0.107
Constant	0.671	0.060	0.209	−0.409
	0.511	0.521	0.532	0.540
R2	0.614	0.641	0.629	0.656
p	0.000	0.000	0.000	0.000
F	17.42	18.33	17.36	18.38

*** $p<0.001$, ** $p<0.01$, * $p<0.05$, + $p<0.1$

Table 3 The empirical results of the mediating role of knowledge acquisition efficiency between commitment-oriented HRM and BMI

	Knowledge acquisition efficiency		Business model innovation			
	Model 1	*Model 2*	*Model 3*	*Model 4*	*Model 5*	*Model 6*
Knowledge acquisition				0.179**	0.140*	0.137*
efficiency				0.065	0.064	0.063
Commitment-oriented HRM		0.163*			0.176**	0.177**
		0.063			0.055	0.054
Collaboration-oriented HRM		0.013				0.152**
		0.066				0.055
CEO educational degree	0.127+	0.117+	0.095	0.072	0.066	0.070
	0.071	0.070	0.062	0.061	0.060	0.059
Firm age	0.010	0.008	0.029+	0.027+	0.025	0.027+
	0.018	0.018	0.016	0.016	0.015	0.015
Firm size	−0.074	−0.087	−0.033	−0.019	−0.035	−0.048
	0.087	0.086	0.076	0.075	0.073	0.072
Founding team member	−0.028	−0.023	−0.016	−0.011	−0.007	−0.005
	0.049	0.048	0.043	0.042	0.041	0.040
Startup experience	0.038	0.021	0.009	0.002	−0.014	−0.009
	0.076	0.075	0.066	0.065	0.064	0.063
Industry experience	0.003	0.005	−0.007	−0.008	−0.006	−0.005
	0.011	0.011	0.010	0.009	0.009	0.009
Teamwork capability	0.076	0.080	0.193***	0.179**	0.187***	0.178**
	0.060	0.059	0.053	0.052	0.051	0.050
Relational capability	−0.018	−0.023	0.080	0.084	0.078	0.076
	0.067	0.066	0.059	0.058	0.056	0.055
Marketing capability	0.003	0.006	0.006	0.006	0.009	0.009
	0.053	0.053	0.047	0.046	0.045	0.044
Design capability	−0.005	−0.014	0.039	0.039	0.029	0.026
	0.064	0.063	0.056	0.055	0.054	0.053
Innovative capability	−0.008	−0.013	0.019	0.021	0.015	0.013
	0.019	0.019	0.016	0.016	0.016	0.015
Firm growth	0.262**	0.236*	0.389***	0.342***	0.331***	0.271**
	0.093	0.096	0.081	0.082	0.080	0.081
Flow-facilitating HRM	−0.030	−0.036	0.392***	0.398***	0.389***	0.397***
	0.066	0.066	0.058	0.057	0.056	0.055
Market uncertainty	−0.002	0.011	−0.072	−0.072	−0.058	−0.053
	0.053	0.053	0.047	0.046	0.045	0.044
Industrial knowledge intensity	0.098	0.085	−0.000	−0.018	−0.028	−0.031
	0.062	0.061	0.054	0.053	0.052	0.051
High-technology industry	0.028	0.045	−0.269*	−0.274*	−0.259*	−0.216*
	0.126	0.126	0.111	0.109	0.106	0.105
Constant	2.562***	2.020**	0.671	0.211	−0.228	−0.687
	0.582	0.640	0.511	0.528	0.533	0.549
R2	0.167	0.198	0.614	0.631	0.651	0.665
p	0.006	0.002	0.000	0.000	0.000	0.000
F	-	-	17.42	17.47	17.94	18.03

*** $p<0.001$, ** $p<0.01$, * $p<0.05$, + $p<0.1$

To test the moderating effect of knowledge application on the relationship between a collaboration-oriented HRM system and BMI, we first treated knowledge application as the dependent variable in the regression models, then BMI. The results are shown in Table 4. In Models 1–3, we tested the effects of a collaboration-oriented HRM system on knowledge application. Model 1 included all the control variables. Model 2 tested the effects of collaboration-oriented HRM on knowledge application; the results suggest a positive and significant effect (b = 0.177, p = 0.023). In Model 3, we included both collaboration- and commitment-oriented HRM systems and found a consistent result (b = 0.179, p = 0.019). Therefore, we may conclude that a collaboration-oriented HRM system positively and significantly affects knowledge application.

In Models 4–7, we tested the effects of a collaboration-oriented HRM system and knowledge application on BMI. Model 4 included all control variables. In Model 5, we added the mediating variable, knowledge application. The empirical results indicated a positive and significant effect of knowledge application (b = 0.194, p = 0.001) on BMI. In Model 6, we included both the mediating variable, knowledge application, and the independent variable, collaboration-oriented HRM system. We found a positive and significant effect of knowledge application (b = 0.174, p = 0.002) and the collaboration-oriented HRM system (b = 0.121, p = 0.036) on BMI. In Model 7, we also added the commitment-oriented HRM system and found a result consistent with that of Model 6. In conclusion, the results support Hypothesis 2.

Test for moderation

To test the moderating effect of flow-facilitating HRM practices in the relationship between BMI and innovation performance, we first tested the effect of BMI on innovation performance and then tested the moderating effect of flow-facilitating HRM practices. In Table 5, we present the results of hierarchical regression (Models 1 and 2), robust regression (Models 3 and 4), and Poisson regression (Models 5 and 6). In Models 1–6, we used the count of new product introductions to measure firm innovation performance. In Model 1, we included all the control variables. We added the variable BMI to Model 2, and the hierarchical regression results indicated a positive and significant effect of BMI (b = 0.647, p = 0.046) on innovation performance. In Model 4, the robust regression results also indicate a positive and significant effect of BMI (b = 0.7967, p = 0.014) on innovation performance. Consistently, in Model 6, the Poisson regression results indicate a positive and significant effect of BMI (b = 0.234, p = 0.001) on innovation performance.

In Table 6, we used an alternative measure comprising three items suggested by Frishammar and Hörte (2005) and Miller and Friesen (1982) to capture innovation performance: the emphasis on R&D, the introduction of new products over time, and changes in products. The results of the hierarchical regression in Model 2 indicate a positive and significant effect of BMI (b = 0.219, p = 0.038) on innovation performance. Consistently, the results of the robust regression in Model 4 also suggest consistent support (b = 0.259, p = 0.021). Therefore, we can conclude that BMI is positively and significantly associated with innovation performance.

Table 4 The empirical results of the mediating role of knowledge application in the relationship between collaboration- oriented HRM and BMI

	Knowledge application			Business model innovation			
	Model 1	*Model 2*	*Model 3*	*Model 4*	*Model 5*	*Model 6*	*Model 7*
Knowledge application					0.194**	0.174**	0.138*
					0.055	0.055	0.055
Commitment-oriented HRM			0.204**				0.129*
			0.072				0.056
Collaboration-oriented HRM		0.177*	0.179*			0.121*	0.171**
		0.077	0.075			0.057	0.054
CEO educational degree	0.103	0.108	0.095	0.095	0.075	0.081	0.073
	0.083	0.082	0.080	0.062	0.060	0.060	0.058
Firm age	0.057**	0.060**	0.057**	0.029+	0.017	0.020	0.020
	0.021	0.021	0.021	0.016	0.016	0.016	0.015
Firm size	0.136	0.121	0.107	−0.033	−0.059	−0.067	−0.075
	0.101	0.100	0.099	0.076	0.074	0.074	0.072
Founding team member	0.022	0.024	0.030	−0.016	−0.021	−0.019	−0.013
	0.057	0.056	0.055	0.043	0.041	0.041	0.040
Startup experience	−0.029	−0.023	−0.045	0.009	0.015	0.019	−0.000
	0.088	0.087	0.086	0.066	0.064	0.064	0.062
Industry experience	0.005	0.006	0.008	−0.007	−0.008	−0.007	−0.006
	0.013	0.013	0.013	0.010	0.009	0.009	0.009
Teamwork capability	0.071	0.060	0.065	0.193***	0.179**	0.173**	0.180***
	0.070	0.069	0.068	0.053	0.051	0.051	0.049
Relational capability	0.135+	0.133+	0.127+	0.080	0.054	0.055	0.055
	0.078	0.077	0.076	0.059	0.057	0.057	0.055
Marketing capability	0.027	0.026	0.031	0.006	0.001	0.001	0.006
	0.062	0.061	0.060	0.047	0.045	0.045	0.044
Design capability	−0.061	−0.064	−0.076	0.039	0.050	0.047	0.035
	0.075	0.074	0.073	0.056	0.055	0.054	0.053
Innovative capability	0.081***	0.078***	0.072**	0.019	0.003	0.003	0.001
	0.022	0.022	0.021	0.016	0.016	0.016	0.016
Firm growth	0.095	0.024	−0.001	0.389***	0.371***	0.324***	0.303***
	0.108	0.111	0.110	0.081	0.079	0.081	0.079
Flow-facilitating HRM	−0.140+	−0.131+	−0.140+	0.392***	0.420***	0.423***	0.411***

Table 4 (continued)

	Knowledge application			Business model innovation			
	0.078	0.077	0.075	0.058	0.057	0.056	0.055
Market uncertainty	0.012	0.018	0.034	−0.072	−0.075	−0.071	−0.056
	0.062	0.061	0.061	0.047	0.045	0.045	0.044
Industrial knowledge	0.131+	0.126+	0.110	−0.000	−0.026	−0.026	−0.035
intensity	0.072	0.071	0.070	0.054	0.053	0.052	0.051
High-technology industry	0.023	0.073	0.089	−0.269*	−0.273*	−0.239*	−0.222*
	0.148	0.147	0.145	0.111	0.107	0.108	0.105
Constant	1.198+	0.658	0.027	0.671	0.438	0.094	−0.413
	0.679	0.711	0.732	0.511	0.499	0.520	0.532
R2	0.273	0.295	0.325	0.614	0.640	0.649	0.668
p	0.000	0.000	0.000	0.000	0.000	0.000	0.000
F	4.11	4.28	4.63	17.42	18.21	17.79	18.27

*** $p<0.001$, ** $p<0.01$, * $p<0.05$, + $p<0.1$

Based on this positive and significant relationship, we moved forward to test the moderating effect of flow-facilitating HRM practices. We first took the count number of new product introductions, as presented in the results of the hierarchical (Models 1–3) and robust (Models 4–6) regressions in Table 7. In Model 3, we tested the moderating effect of flow-facilitating HRM practices and found a positive and significant effect (b = 0.337, $p=0.039$) on innovation performance. We also tested its moderating effect in Model 6, with the results indicating a positive and significant moderating effect (b = 0.390, $p=0.014$) of flow-facilitating HRM practices on the relationship between BMI and innovation performance. We used an alternative measure of innovation performance (three items assessing emphasis on R&D, the introduction of new products over time, and changes in products) and further examined the moderating effect of flow-facilitating HRM practices on the relationship between BMI and innovation performance. Figures 2 and 3 present graphical evidence of the moderating effect of flow-facilitating HRM practices. We found consistent evidence that supports Hypothesis 3.

Robustness check

To ensure the validity of our results, we conducted a robustness check using Partial Least Squares Structural Equation Modeling (PLS-SEM). We examined the proposed hypotheses using a bootstrapping technique with 5000 sub-samples (see Fig. 4; Table 8). The H1 was supported ($\beta=0.082$; $p<0.01$), confirming that commitment-oriented HRM (COM) positively affects the BMI of SMEs, with knowledge acquisition efficiency (KAE) mediating this effect. The H2 was also supported ($\beta=0.086$; $p<0.01$), indicating that collaboration-oriented HRM (COL) positively affects the BMI of SMEs, with knowledge acquisition (KA) as the mediating variable. Furthermore, the H3 was supported ($\beta=0.134$; $p<0.01$), showing that flexible, fit-for-purpose HRM

Table 5 Empirical results of the effects of BMI on innovation performance

	Innovation performance		Innovation performance		Innovation performance	
	Model 1	*Model 2*	*Model 3*	*Model 4*	*Model 5*	*Model 6*
BMI	-	0.647*	-	0.796*	-	0.234**
	-	0.322	-	0.320	-	0.073
Knowledge application	0.518*	0.437+	0.495*	0.441+	0.160***	0.131**
	0.235	0.236	0.234	0.234	0.046	0.047
Knowledge acquisition	0.506+	0.429	0.395	0.306	0.141**	0.112*
efficiency	0.269	0.269	0.268	0.267	0.050	0.051
Commitment-oriented HRM	0.283	0.183	0.418+	0.322	0.070	0.029
	0.231	0.234	0.230	0.232	0.047	0.048
Collaboration-oriented HRM	0.105	0.021	0.277	0.196	0.012	−0.011
	0.236	0.238	0.236	0.236	0.046	0.047
CEO educational degree	0.224	0.184	0.215	0.149	0.086+	0.066
	0.250	0.248	0.249	0.246	0.047	0.047
Firm age	0.032	0.019	0.059	0.036	0.010	0.006
	0.066	0.065	0.065	0.065	0.011	0.011
	0.626	0.772	0.372	0.577	0.358	0.580
Firm size	−0.103	−0.062	−0.231	−0.161	−0.015	0.006
	0.306	0.304	0.305	0.301	0.060	0.061
Founding team member	0.209	0.215	0.151	0.187	0.104**	0.102**
	0.170	0.169	0.170	0.167	0.033	0.033
Startup experience	−0.104	−0.101	−0.074	−0.158	−0.023	−0.032
	0.265	0.263	0.265	0.261	0.047	0.047
Industry experience	0.022	0.027	0.023	0.040	0.004	0.006
	0.039	0.039	0.039	0.039	0.007	0.007
Teamwork capability	0.034	−0.076	0.120	−0.035	0.023	−0.010
	0.212	0.217	0.211	0.215	0.043	0.045
Relational capability	−0.036	−0.075	−0.239	−0.343	−0.018	−0.031
	0.235	0.234	0.235	0.232	0.045	0.045
Marketing capability	0.111	0.108	0.251	0.273	0.039	0.034
	0.186	0.185	0.186	0.183	0.037	0.037
Design capability	−0.005	−0.028	−0.003	−0.057	−0.006	−0.005
	0.225	0.223	0.225	0.221	0.045	0.045
Innovative capability	0.148*	0.146*	0.176*	0.172*	0.029*	0.027*
	0.069	0.068	0.069	0.068	0.012	0.012
	0.033	0.035	0.012	0.012	0.020	0.032

Table 5 (continued)

	Innovation performance		Innovation performance		Innovation performance	
Firm growth	−0.224	−0.403	−0.187	−0.395	−0.073	−0.126+
	0.343	0.352	0.343	0.349	0.066	0.068
Flow-facilitating HRM	1.140***	0.872**	1.021***	0.695**	0.327***	0.228***
	0.235	0.268	0.234	0.266	0.046	0.055
Market uncertainty	−0.049	−0.012	−0.052	−0.017	−0.001	0.022
	0.187	0.186	0.187	0.185	0.038	0.039
Industrial knowledge intensity	−0.258	−0.230	−0.186	−0.166	−0.050	−0.043
	0.218	0.217	0.218	0.215	0.041	0.041
High-technology industry	0.459	0.606	0.900*	1.046*	0.031	0.062
	0.446	0.448	0.445	0.444	0.091	0.092
Constant	−7.340**	−6.916**	−8.128***	−7.729***	−2.305***	−2.258***
	2.313	2.303	2.309	2.284	0.494	0.503
R2	0.367	0.382			0.203	0.212
p	0.000	0.000	0.000	0.000	0.000	0.000
F/Likelihood	4.96	5.00	5.67	6.06	−461.125	−456.016

*** $p<0.001$, ** $p<0.01$, * $p<0.05$, + $p<0.1$

(FFHRM) positively moderates the relationship between BMI and innovation performance in SMEs. This robustness check reaffirms the strength and reliability of our findings, providing a solid foundation for the conclusions drawn from our study.

Discussion

This research explores the impact of commitment- and collaboration-oriented HRM systems on BMI and subsequent effects on innovation performance. The study finds positive and significant effects of commitment- and collaboration-oriented HRM systems on BMI. It further identifies knowledge acquisition efficiency as a mediating mechanism between commitment-oriented HRM systems and BMI and knowledge application as a mediating mechanism between collaboration-oriented HRM systems and BMI. Moreover, flow-facilitating HRM practices strengthen the positive effects of BMI on innovation performance.

Theoretical implications

This study makes three related contributions to the literature by integrating knowledge management concepts with an examination of HRM practices in the context of BMI. First, this study contributes to the theoretical understanding of BMI by integrating insights from knowledge management theory to uncover how HRM systems enable the knowledge flows required for business model transformation. While prior

Table 6 Empirical results of the effects of BMI on innovation performance

	Innovation performance		Innovation performance	
	Model 1	*Model 2*	*Model 3*	*Model 4*
Business model innovation		0.219*		0.259*
		0.104		0.111
Knowledge application	0.200**	0.172*	0.227**	0.191*
	0.076	0.076	0.081	0.081
Knowledge acquisition efficiency	0.073	0.047	0.065	0.045
	0.087	0.087	0.093	0.093
Commitment oriented-HRM	0.148*	0.114	0.138+	0.104
	0.075	0.076	0.080	0.081
Collaboration oriented-HRM	0.030	0.002	0.036	−0.001
	0.076	0.077	0.082	0.082
CEO educational degree	0.166*	0.152+	0.164+	0.147+
	0.081	0.080	0.086	0.086
Firm age	−0.014	−0.019	−0.011	−0.018
	0.021	0.021	0.022	0.022
Firm size	−0.041	−0.028	−0.064	−0.035
	0.099	0.098	0.106	0.105
Founding team member	0.015	0.018	0.006	0.005
	0.055	0.054	0.059	0.058
Startup experience	0.074	0.075	0.050	0.051
	0.086	0.085	0.092	0.091
Industry experience	0.015	0.016	0.019	0.021
	0.012	0.012	0.013	0.013
Teamwork capability	−0.040	−0.078	−0.035	−0.082
	0.068	0.070	0.073	0.075
Relational capability	−0.080	−0.093	−0.089	−0.117
	0.076	0.076	0.081	0.081
Marketing capability	0.027	0.026	0.021	0.020
	0.060	0.059	0.064	0.063
Design capability	0.007	−0.000	0.005	−0.010
	0.073	0.072	0.078	0.077
Innovative capability	0.029	0.028	0.025	0.026
	0.022	0.022	0.024	0.023
Firm growth	−0.048	−0.108	−0.039	−0.110
	0.111	0.114	0.119	0.121
Flow-facilitating HRM	0.446***	0.355***	0.453***	0.345***
	0.076	0.087	0.081	0.092
Market uncertainty	0.038	0.051	0.041	0.047
	0.060	0.060	0.065	0.064
Industrial knowledge intensity	−0.063	−0.053	−0.052	−0.041
	0.070	0.070	0.075	0.075
High-technology industry	0.041	0.091	0.021	0.083
	0.144	0.145	0.154	0.155
Constant	−0.502	−0.358	−0.513	−0.287
	0.750	0.746	0.802	0.796
R2	0.389	0.405	-	-

Table 6 (continued)

	Innovation performance		Innovation performance	
	Model 1	*Model 2*	*Model 3*	*Model 4*
p	0.000	0.000	0.000	0.000
F	5.46	5.51	4.94	5.13

*** $p<0.001$, ** $p<0.01$, * $p<0.05$, + $p<0.1$

research has established the importance of knowledge management for innovation (e.g., Chen & Huang, 2009; Seeck & Diehl, 2017), this study advances the literature by demonstrating how specific HRM systems operationalize distinct knowledge management processes, namely knowledge acquisition efficiency and knowledge application, in the context of BMI. Commitment-oriented HRM systems are shown to enhance knowledge acquisition by fostering trust, collaboration, and alignment, which create the organizational conditions necessary for gathering and integrating knowledge resources. In contrast, collaboration-oriented HRM systems facilitate the application of external knowledge by enabling firms to integrate insights from external partnerships into BMI efforts. Through positioning HRM systems as mechanisms that embed knowledge management processes within organizational practices, this study provides a theoretical bridge between HRM and knowledge management, showing how these systems directly shape the processes necessary to rethink and redesign business models. This contribution extends the knowledge management perspective in BMI by identifying the specific pathways through which HRM systems influence knowledge flows, thereby strengthening the conceptual link between HRM practices and BMI.

Second, this study makes a significant theoretical contribution by providing empirical evidence of the differentiated and complementary roles of commitment-oriented and collaboration-oriented HRM systems in enabling BMI. While previous studies have conceptually distinguished between these HRM systems (e.g., Zhou et al., 2013; Hong et al., 2019), they have not explored how these systems individually contribute to the knowledge processes critical for BMI. This research empirically demonstrates that commitment-oriented HRM systems primarily support knowledge acquisition by strengthening internal organizational capabilities to systematically acquire and integrate knowledge resources. Conversely, collaboration-oriented HRM systems focus on knowledge application, emphasizing external engagement and adaptability to ensure that external knowledge is effectively adapted and applied in BMI efforts. By empirically validating the distinct roles of these HRM systems, this study challenges the prevailing assumption that HRM systems have a uniform or generalized impact on innovation. Instead, it highlights how their differentiated contributions address the dual demands of BMI: building a stable internal foundation for knowledge acquisition and simultaneously ensuring the external adaptability required for applying knowledge in dynamic business model transformation. This contribution advances the HRM and BMI literature by providing a more precise understanding of how firms can strategically align different HRM systems to support the BMI.

Third, this study contributes to the literature by introducing and empirically validating the role of flow-facilitating HRM practices as critical organizational enablers

Table 7 Empirical results of the moderating effects in the relationship between BMI and innovation performance

	Innovation performance			Innovation performance		
	Model 1	*Model 2*	*Model 3*	*Model 4*	*Model 5*	*Model 6*
Business model innovation	1.169***	0.647*	−0.448	1.241***	0.796**	−0.418
	0.287	0.322	0.617	0.281	0.320	0.599
Flow-facilitating HRM	-	0.872**	−0.588	-	0.695***	−0.970
	-	0.268	0.752	-	0.266	0.730
Business model innovation X	-	-	0.337*	-		0.390**
Flow-facilitating HRM	-	-	0.162	-		0.157
Knowledge application	0.283	0.437+	0.454+	0.307	0.441+	0.496*
	0.238	0.236	0.234	0.232	0.234	0.227
Knowledge acquisition efficiency	0.347	0.429	0.376	0.310	0.306	0.216
	0.209	0.113	0.163	0.251	0.253	0.406
Commitment oriented-HRM	0.148	0.183	0.112	0.261	0.322	0.222
	0.240	0.234	0.234	0.235	0.232	0.228
Collaboration oriented-HRM	−0.063	0.021	0.004	0.130	0.196	0.161
	0.243	0.238	0.236	0.237	0.236	0.229
CEO educational degree	0.192	0.184	0.176	0.130	0.149	0.157
	0.255	0.248	0.246	0.249	0.246	0.239
Firm age	0.033	0.019	0.009	0.049	0.036	0.014
	0.067	0.065	0.065	0.065	0.065	0.063
Firm size	−0.047	−0.062	0.026	−0.092	−0.161	−0.064
	0.312	0.304	0.304	0.305	0.301	0.295
Founding team member	0.289+	0.215	0.249	0.269	0.187	0.215
	0.172	0.169	0.168	0.168	0.167	0.163
Startup experience	−0.204	−0.101	−0.086	−0.316	−0.158	−0.166
	0.268	0.263	0.261	0.262	0.261	0.253
Industry experience	0.047	0.027	0.016	0.060	0.040	0.034
	0.040	0.039	0.039	0.039	0.039	0.038
Teamwork capability	−0.091	−0.076	−0.074	−0.064	−0.035	−0.054
	0.223	0.217	0.215	0.218	0.215	0.209
Relational capability	−0.129	−0.075	−0.091	−0.390+	−0.343	−0.357
	0.240	0.234	0.232	0.234	0.232	0.225
Marketing capability	0.063	0.108	0.091	0.242	0.273	0.252
	0.189	0.185	0.183	0.185	0.183	0.178
Design capability	−0.028	−0.028	0.013	−0.092	−0.057	−0.032
	0.229	0.223	0.222	0.224	0.221	0.216
Innovative capability	0.214**	0.146*	0.102	0.224***	0.172**	0.107
	0.067	0.068	0.071	0.065	0.068	0.069
Firm growth	−0.406	−0.403	−0.387	−0.449	−0.395	−0.365
	0.362	0.352	0.349	0.353	0.349	0.339
Market uncertainty	0.014	−0.012	0.068	−0.020	−0.017	0.088

Table 7 (continued)

	Innovation performance			Innovation performance		
	Model 1	*Model 2*	*Model 3*	*Model 4*	*Model 5*	*Model 6*
	0.191	0.186	0.189	0.187	0.185	0.183
Industrial knowledge intensity	−0.087	−0.230	−0.251	−0.045	−0.166	−0.201
	0.218	0.217	0.215	0.213	0.215	0.209
High-technology industry	0.747	0.606	0.424	1.064*	1.046*	0.758+
	0.458	0.448	0.452	0.447	0.444	0.439
Constant	−5.659**	−6.916***	−2.266	−6.733**	−7.729***	−2.273
	2.332	2.303	3.196	2.278	2.284	3.106
R2	0.343	0.382	0.397			
p	0.000	0.000	0.000	0.000	0.000	0.000
F	4.47	5.00	5.06	5.83	6.06	6.48

*** $p<0.001$, ** $p<0.01$, * $p<0.05$, + $p<0.1$

Fig. 2 Moderating effect of flow-facilitating HRM practices (innovation performance measured by the number of new products)

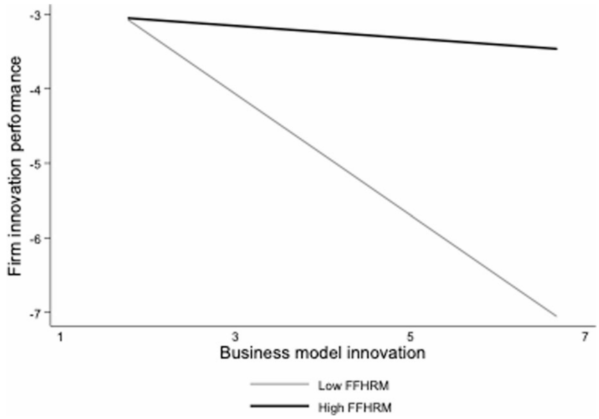
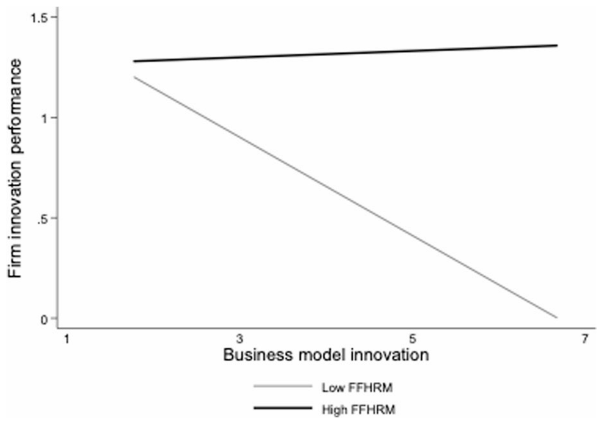


Fig. 3 Moderating effect of flow-facilitating HRM practices (innovation performance measured by three scale items)



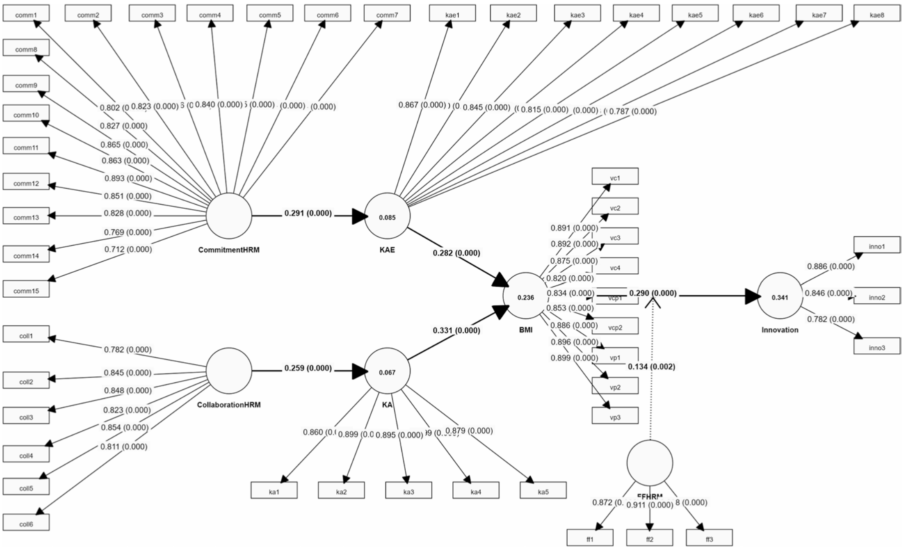


Fig. 4 PLS path model

Table 8 Hypothesis Testing of PLS-SEM

Hypotheses	Relationship	Path coefficient	Standard deviation	T statistics	*P* value	Results
Baseline tests						
	COM -> KAE	0.291	0.063	4.610	0.000	-
	COL -> KA	0.259	0.066	3.920	0.000	-
	KAE -> BMI	0.282	0.061	4.649	0.000	-
	KA -> BMI	0.331	0.060	5.479	0.000	-
	FFHRM -> INN	0.320	0.074	4.304	0.000	-
	BMI -> INN	0.290	0.079	3.690	0.000	-
Mediation						
H1	COM -> KAE -> BMI	0.082	0.028	2.893	0.004	Accepted**
H2	COL -> KA -> BMI	0.086	0.032	2.699	0.007	Accepted**
Moderation						
H3	FFHRM * BMI -> INN	0.134	0.043	3.089	0.002	Accepted**

*** $p<0.001$, ** $p<0.01$, * $p<0.05$

of BMI implementation. While much of the existing research on BMI has focused on macro-level factors—such as organizational structures, technological capabilities, or industry dynamics—that influence BMI outcomes (e.g., Foss & Saebi, 2017; Amit & Zott, 2012), this study shifts the focus to HRM practices as micro-level enablers that facilitate the transition from BMI design to successful implementation. Flow-facilitating HRM practices, such as effective communication, cross-functional coordination, and employee participation, are shown to create the conditions necessary for employees to actively engage with BMI processes. These practices ensure seamless knowledge flows across organizational levels and functions, enabling firms to translate innovative business models into tangible outcomes, such as new product or service development. By addressing the underexplored role of HRM practices in BMI implementation, this study extends the BMI literature by demonstrating that successful BMI implementation depends not only on the conceptual redesign of business models but also on HRM systems' ability to operationalize and embed BMI within the organization. This contribution highlights the importance of aligning HRM practices with BMI implementation to overcome the challenges of translating strategic innovation into actionable performance improvements.

Practical implications

This study provides several practical suggestions for managers in SMEs to implement HRM practices that can support BMI. These suggestions focus on specific approaches managers might consider to foster employee engagement, integrate external expertise, and coordinate innovation efforts effectively.

First, managers might consider implementing commitment-oriented HRM practices to build a motivated and cohesive workforce that actively contributes to BMI efforts. Practices that emphasize trust, mutual respect, and long-term employee development could create an environment conducive to innovation. For example, providing employees with clear communication about organizational goals, recognizing their contributions, and offering opportunities for professional growth could strengthen their engagement in BMI initiatives. Additionally, managers might organize innovation workshops, brainstorming sessions, or team meetings to encourage employees to share knowledge, collaborate on solving organizational challenges, and identify new opportunities for business model transformation. Such practices can help employees feel valued and aligned with the organization's innovation objectives.

Second, collaboration-oriented HRM practices may be useful for SMEs seeking to build external connections and integrate external knowledge into their innovation processes. Managers could explore establishing formal partnerships with academic institutions, industry associations, or external experts to facilitate knowledge exchange and collaboration. For instance, joint workshops, training programs, or research projects with external stakeholders could provide employees with access to new insights, tools, and expertise that support BMI. Similarly, participating in professional networks or industry forums may enable managers to identify successful external practices and adapt them to their own organizations. Managers might also initiate co-development projects with external partners, such as technology firms or suppliers, to explore new opportunities for business model transformation. These practices could help SMEs broaden their knowledge base and respond more effectively to changes in their markets or industries.

Third, aligning HRM practices with specific knowledge processes could help managers maximize the impact of these systems on BMI. Commitment-oriented HRM practices might focus on improving knowledge acquisition and sharing, while collaboration-oriented HRM practices could emphasize the effective application of external knowledge. Managers may find it useful to monitor whether their HRM practices are achieving these goals. For instance, they might evaluate the effectiveness of knowledge-sharing systems by tracking participation in collaborative activities or assessing whether employees are integrating shared knowledge into their work. Similarly, managers could examine how insights gained from external collaborations are being applied to BMI projects and whether these efforts are generating tangible outcomes, such as new product or service development. Conducting periodic reviews of HRM practices and their alignment with knowledge processes might help managers identify areas for improvement and adapt their strategies to better support BMI.

Finally, managers might consider adopting flow-facilitating HRM practices to ensure effective coordination and sustained engagement across their organizations during BMI implementation. Clearly communicating the objectives and expected outcomes of BMI initiatives to all employees could help foster a shared understanding of the innovation process. Encouraging collaboration across departments may also support more integrated and innovative solutions. Managers might create cross-functional teams to address specific BMI challenges, allowing employees from different areas of the organization to share their expertise and perspectives. Additionally, providing feedback and recognizing employees' contributions to BMI efforts could help sustain motivation and engagement. For instance, celebrating milestones or offering rewards for innovative ideas might reinforce a culture of innovation and encourage employees to remain actively involved in BMI initiatives.

Limitations and future research

As with any study, this research has several limitations that warrant consideration and provide opportunities for future investigation. First, the sample firms in this study consist of high-tech SMEs engaged in innovation activities, which provided an ideal research setting for examining the effects of HRM systems on BMI. This focus also helps define the study's boundary conditions within the SME context, a domain recognized as deserving significant scholarly attention in the field of BMI (Foss & Saebi, 2017). However, the generalizability of the findings beyond this specific context remains uncertain. Future research could address this limitation by investigating whether the results hold in alternative contexts, such as non-high-tech sectors, large organizations, or diverse business environments across different countries. These extensions are particularly important, as the design and implementation of HRM systems are likely to be shaped by cultural, institutional, and structural contingencies.

Second, this study identified two distinct mediating mechanisms, knowledge acquisition efficiency and knowledge application efficiency, from the perspective of organizational knowledge management. While these mechanisms provide valuable insights, they do not capture the full spectrum of processes through which HRM systems influence BMI. Future research could explore additional mediating mechanisms, such as employee-level psychological factors (e.g., motivation, engagement, or trust) or firm-level processes (e.g., organizational climate, leadership, or structural design). These complementary perspectives could enrich our understanding of how HRM systems contribute to BMI by uncovering additional pathways beyond knowledge management.

Third, while this study focused on the distinct effects of commitment- and collaboration-oriented HRM systems, we did not propose or examine interactions between them. This decision was made to reduce the study's complexity and maintain a clear focus on the unique contributions of each HRM system to knowledge management processes. However, we acknowledge that potential synergies between these systems may exist. Future research could investigate whether and how commitment- and collaboration-oriented HRM systems reinforce or complement one another in shaping knowledge management processes and BMI outcomes. For instance, future studies could examine how the simultaneous implementation of these HRM systems impacts innovation performance, using longitudinal or large-scale datasets to capture the dynamic nature of their interaction.

Fourth, the theoretical framework of this study focused on the dominant pathways: commitment-oriented HRM systems that enhance knowledge acquisition efficiency, and collaboration-oriented HRM systems that improve knowledge application efficiency. This focus was designed to ensure parsimony and conceptual clarity. However, potential cross-path relationships, such as the influence of commitment-oriented HRM systems on knowledge application and of collaboration-oriented HRM systems on knowledge acquisition, were not examined. These relationships may operate through more complex or indirect mechanisms, possibly moderated by organizational contexts or mediated by other processes. Future research could address this limitation by exploring cross-path relationships and their underlying dynamics through more specific research designs, such as multilevel studies or longitudinal designs, to provide deeper insights into these interactions.

Finally, this study primarily emphasized mediating mechanisms from a knowledge management perspective, while potential contingent factors that may influence the relationship between HRM systems and BMI were not investigated. To address this limitation, future studies could identify and test moderators at various levels of analysis. For example, firm-level moderators, such as organizational values, culture, and structural design, or micro-level moderators, such as characteristics of human capital (e.g., skills, expertise, diversity), could provide important insights into the contextual conditions under which HRM systems influence BMI. Exploring these moderators would allow for a richer understanding of the boundary conditions of HRM systems' effectiveness.

Conclusion

To unpack the relationship between HRM systems and innovation, this study extends existing knowledge by examining the different effects of internal commitment- and external collaboration-oriented HRM systems on BMI in SMEs. These two HRM systems represent highly innovative work systems that emphasize distinctive HRM patterns (Zhou et al., 2013). We proposed that these two systems would influence the innovative design of business models, which drive SMEs to create, deliver, and capture value for/from their customers. We found a positive relationship between internal commitment-oriented HRM systems and BMI, and that the relationship is mediated by knowledge acquisition efficiency. We also found a positive relationship between externally collaborative-oriented HRM systems and BMI, which is mediated by knowledge application. In addition, we identified a positive relationship between BMI and innovation performance (i.e., new product development) and that this relationship is strengthened by flow-facilitating HRM practices. In short, the findings of this study indicate that a highly commitment-oriented system enhances the efficiency of knowledge acquisition, in turn producing a high BMI, while a highly collaboration-oriented system facilitates knowledge application, also producing a high BMI. Thus, SMEs can achieve high innovation performance in new product development by implementing a high BMI.

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